Modeling the Diffusion of Fake News in Social Media

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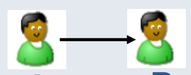
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Background

Online Social Networks

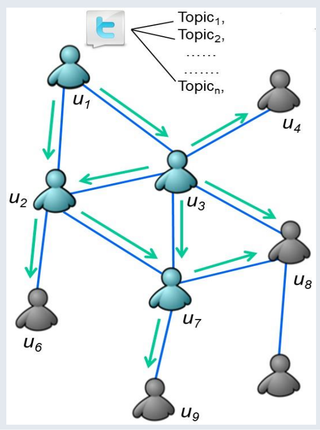
- A **group** of users who are **closely connected** to each other.
- **Interact** with each other through different activities like posting and sharing news, opinions etc.





Attributed Graph

- **Rich** attributed nodes (users) with properties.



Modeling Diffusion of Fake News

- To model and characterize ***fake news*** sharing through different levels of **influential factors:**
  - users
  - network
  - news
- Predicting fake news sharing is considered as **classification** task.

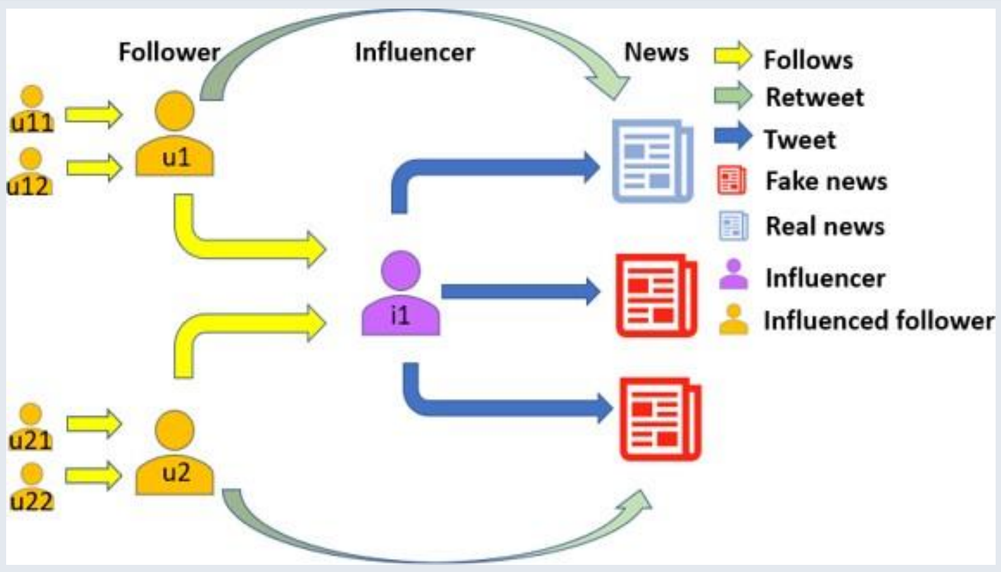
Literature Review

- Shu et al. [1] focused on considering a **tri-relationship** among user, publisher, and news content.
- Yaqub et al [2] proposed method that dampen and reduce the spread of misinformation by annotating it with **credibility indicators**.
- Giachanou et al. [3] proposed an approach based on a **convolutional neural network** to process the user Twitter feed in combination with **features** representing user personality traits and linguistic patterns used in their tweets to address the problem of discriminating between fake news spreaders and fact-checkers.

Limitation:

- Paid less focus on the influence level of neighbors.

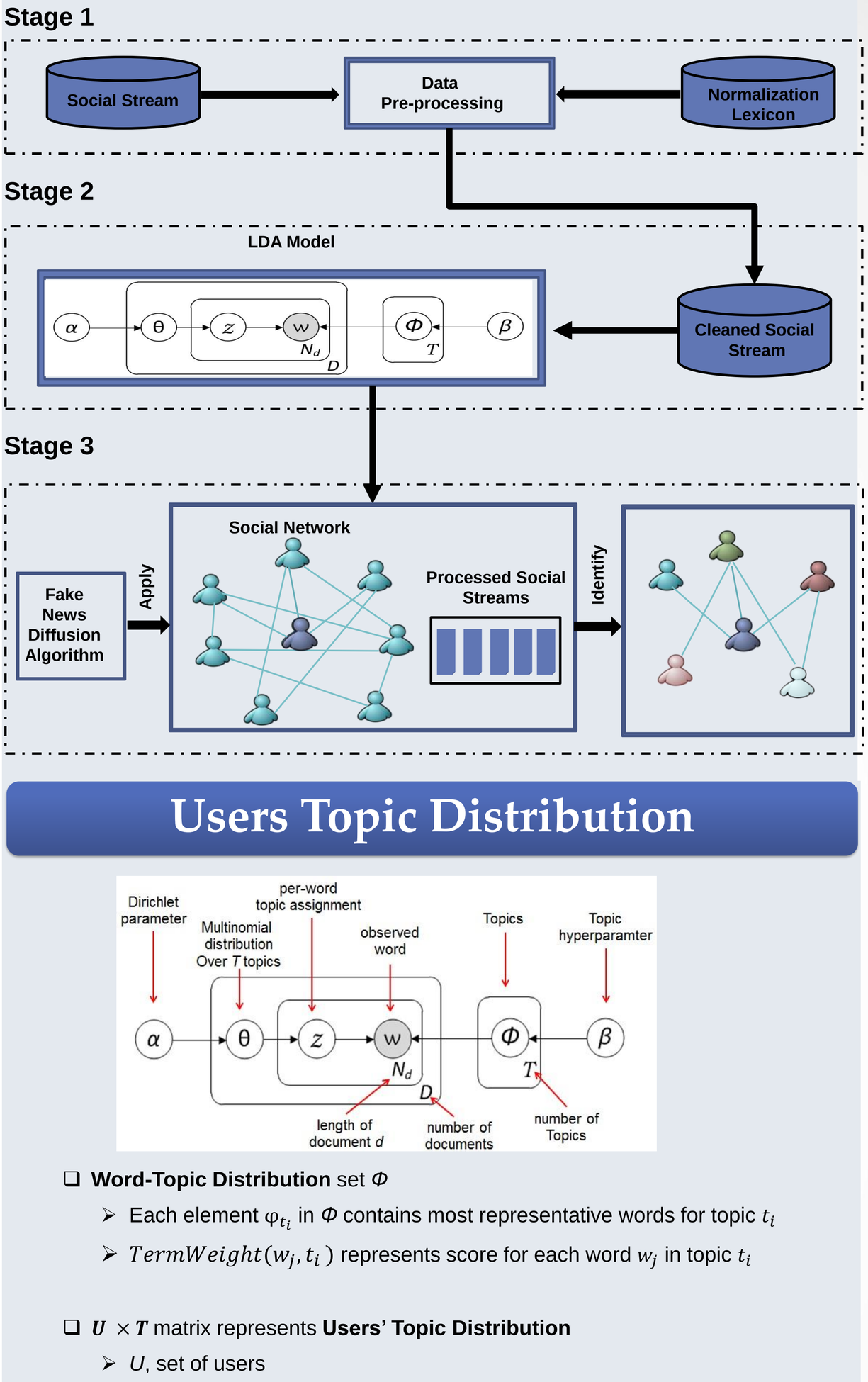
Problem Formulation



Problem Formulation

- Given, Social Graph ***G = (U, E, N)***
  - *U*, set of users as *nodes*
  - *E*, set of directed edges representing *social connections* (relationships)
  - *N*, set of *news*
- For any user ***u*** is **influenced** on a news item (real or fake) ***n*** by her **neighbours** (connected users)
  - Predict whether ***u*** will share news item ***n*** among her connections
- **Features** to be considered:
  - User-based features
    - Demographics, Profile information, Emotions, Personality
  - Network-based features
    - User Centrality, Tie Strength
  - News-based features
    - Stylistic Features, Complexity, and Psychological aspects

Proposed Methodology



Users Topic Distribution

- **Word-Topic Distribution** set Φ
  - Each element φ_{t_i} in Φ contains most representative words for topic t_i
  - $TermWeight(w_j, t_i)$ represents score for each word w_j in topic t_i
- $U \times T$ matrix represents **Users' Topic Distribution**
  - *U*, set of users
  - *T*, number of topics
  - Each row represents topic distribution θ_u for each user u
  - Each cell represents specific topic distribution for a user

Data Set and Expected Result

Date Sets: From(https://github.com/KaiDMML/FakeNewsNet)

# Users	# Fake News	# Shared	# Not Shared
1,557	169	527	7,134

- We like to **compare** various **machine learning** algorithms, namely Logistic Regression, Support Vector Machine (SVM), Random Forest etc.
- As evaluation metrics, we like to consider the **Area Under the ROC Curve** (AUROC) and **Average Precision** (AvgP)

References

1. W. Yaqub, O. Kakhidze, Morgan L Brockman, Nasir Memon, and Sameer Patil. Effects of credibility indicators on social media news sharing intent. In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, pp. 1–14, 2020.
2. Kai Shu, Suhang Wang, and Huan Liu. Beyond news contents: The role of social context for fake news detection. In Proceedings of the Twelfth ACM International Conference on Web Search and Data Mining, pp. 312–320, 2019.
3. A. Giachanou, E. A. Rissola, B. Ghanem, F. Crestani, and P. Rosso, "The role of personality and linguistic patterns in discriminating between fake news spreaders and fact checkers," in NLDB 2020, pp. 181--192.

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